Yale JACKSON SCHOOL OF GLOBAL AFFAIRS

Assignment 3: The Effects of Working from Home Due Sunday, March 1

GLBL 5021 Applied Methods of Analysis II

Learning goals:

i) conduct data analysis using data from an experiment (create balance tables, visualization, and run a regression)
ii) write-up interpretation from experiment results
iii) start brainstorming data for the final project

Question of interest: Does working from home improve productivity? We will replicate results from the paper by Bloom et al. (2014) and go through some analysis that one should do when using RCT data (a cleaner version coming from the textbook).

Download "HW3_wfh_start" code from canvas in your preferred language. The code provides you with a start and some example codes to help you. Continue the code while following the instructions below. Once you finish your code, copy and paste it at the end of your submission. Submit the pdf of your assignment on Gradescope (canvas).

Part I: Checking for Balance using Table of Descriptive Statistics

From the potential outcome model lecture, we learned that conditional independence, $E[Y^0|D=1]-E[Y^0|D=0]=0$, needs to hold. In practice—since we don't have data from two dimensions—researchers check for this in experimental data by constructing "balance tables". Balance tables are descriptive statistics of the control and treatment group that check whether characteristics are significantly different or not. Appendix Table A.2 (check page 50 in the paper). You will replicate this table and understand where the numbers come from.

1.1 Creating a balance table is quite easy. You need the average of variable "x" by group and then determine whether the difference is statistically significant. To demonstrate:

- Write a code to get the average age in control and then in treatment.
- Another way to get average difference and do hypothesis testing on whether the difference is 0: Run a regression of age on treatment (make sure use robust!). Check the p-value on the coefficient of treatment. This p-value should be the same as the one on the Appendix Table A.2.
- Explain whether your results are the same as the ones on the table for the variable age. You can try another variable to be sure.

Graph and/or written answers required.

1.2 To create a balance table, follow the supplemental code HW3_example_codes. For R, this is easy because you will use the package datasummary_balance() from the package "model_summary". In python, the example code makes you do this manually, but the logic is quite easy. Your balance

tables should include all variables EXCEPT for: personid, treatment, perform11 (this is performance after treatment), quitjob, and phonecalls1. The latter two are outcomes of interest after the treatment so we don't check for balance. Include your table below.

Graph and/or written answers required.

1.3 Provide a paragraph write-up on how different the samples in the control and treatment group are. Note any variables where the difference is statistically significant.

Graph and/or written answers required.

Part II: Visualizing Group Differences

You have created scatterplot graphs for previous homework. For this one, let's create a bar graph since you wanted to show and investigate how much quitjob differ between the two groups.

2.1 Use the help of LLMs or the example code provided to create a bar graph on the share of people who quit on the control versus treatment group. Provide the graph below.

Graph and/or written answers required.

2.2 Based on the graph you created in 2.1, write what you find in two to three sentences.

Graph and/or written answers required.

Part III: Running Regressions to get Average Treatment Effect

3.1 You will now run several regressions to get the ATE. Present the regression results side-by-side. For our purposes, let's do an LPM for now.

- Model 1: Regress quit rates on treatment.
- Model 2: Regress phone calls on treatment. Use the sample where ordertaker=1. These are the only people who are on the phone.
- Model 3: Regress quit rates on treatment + add control variables that were significantly different from part 1.2.
- Model 4: Regress phone calls on treatment + add control variables that were significantly different from part 1.2. Use the sample where ordertaker=1.

Note: phonecalls1 is the total number of phone calls in thousands during experiment period.

3.2 Write a paragraph interpreting the results. We always add controls from our balance tables that are not significant to ensure we still have conditional independence. Does adding these controls change the results?

Graph and/or written answers required.

3.3 For fun, let's replicate the results from Table VIII (page 40) using a probit model instead of an LPM. Run one regression of jobquits on treatment, age, men, married, and cost of commuting (Model I only) using LPM and probit. Check the marginal effects of the probit and compare them with the LPM. Calculate the Brier Score to see which one performs better. Write how the model compares.

Graph and/or written answers required.

3.4 Write a few sentences about the key results (coefficient/marginal effects) found in 3.3.

Graph and/or written answers required.

Part IV: Brainstorming Data for the Final Project

This section helps you get started thinking about the final project. You will also receive some feedback on the feasibility of the project.

4.1 Write down three potential questions that you can explore.

Graph and/or written answers required.

4.2 Find a dataset and list variables from the dataset that can be used to answer the questions above. Explain why the variables are helpful.

Graph and/or written answers required.

4.3 Explain how the dataset can be used to answer the question (what methodology or data analysis techniques can be used?)

Graph and/or written answers required.